

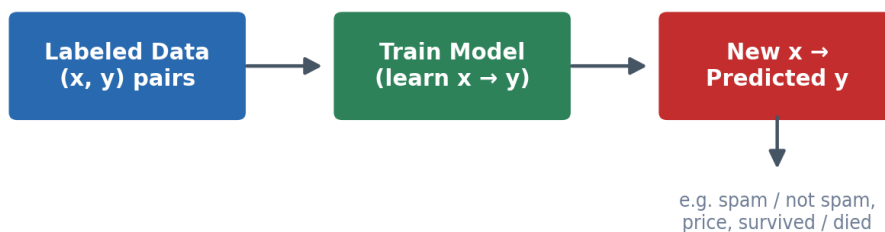
Supervised vs. Unsupervised Learning

An essential-knowledge quick reference

1. Supervised Learning

Supervised learning trains a model on data that already has the answer attached — each example is a pair of inputs and a known correct output. The model's job is to learn the mapping from input to output well enough to predict the output for new, unseen inputs.

Supervised Learning: Learn from Labeled Examples



Definition

Given labeled training data (x, y) , learn a function f such that $f(x) \approx y$, then use f to predict y for new x you haven't seen the answer to.

The two subtypes

- **Classification** — Predicting a category (spam/not spam, survived/died, fraud/legitimate). Algorithms: logistic regression, decision trees, random forest, SVM.
- **Regression** — Predicting a continuous number (house price, temperature, sales forecast). Algorithms: linear regression, random forest regression, XGBoost regression.

Use cases

- Email spam filtering
- Customer churn prediction
- Credit risk / fraud detection
- House price / sales forecasting
- Medical diagnosis from labeled scans or test results

Pros

- Performance is directly measurable — you always know how right or wrong the model is
- Generally more accurate for well-defined prediction tasks, given enough labeled data
- Wide range of mature, well-understood algorithms and evaluation tools

Cons

- Requires labeled data, which is often expensive, slow, or impractical to collect at scale
- Model quality is capped by label quality — biased or noisy labels produce a biased or noisy model
- Doesn't generalize to discovering patterns nobody thought to label in the first place

2. Unsupervised Learning

Unsupervised learning works with data that has no labels at all — just raw inputs. Instead of learning to predict a known answer, the model's job is to find structure, patterns, or groupings that exist in the data on their own.

Unsupervised Learning: Find Structure with No Labels



Definition

Given only unlabeled data x , discover hidden structure — groups, patterns, or a simpler representation — without ever being told what the "right answer" looks like.

The main subtypes

- **Clustering** — Group similar data points together without predefined categories. Algorithms: k-Means, hierarchical clustering, DBSCAN.
- **Dimensionality reduction** — Compress data into fewer dimensions while preserving as much information as possible. Algorithms: PCA, SVD/matrix factorization, t-SNE.
- **Association** — Find items that frequently co-occur (e.g., “customers who bought X also bought Y”). Algorithms: Apriori, FP-Growth.

Use cases

- Customer segmentation for marketing
- Anomaly / fraud detection (flagging what doesn't fit any known pattern)
- Topic discovery in large text collections
- Compressing high-dimensional data for visualization or as input to another model
- Collaborative filtering in recommendation systems (finding latent taste patterns via matrix factorization)

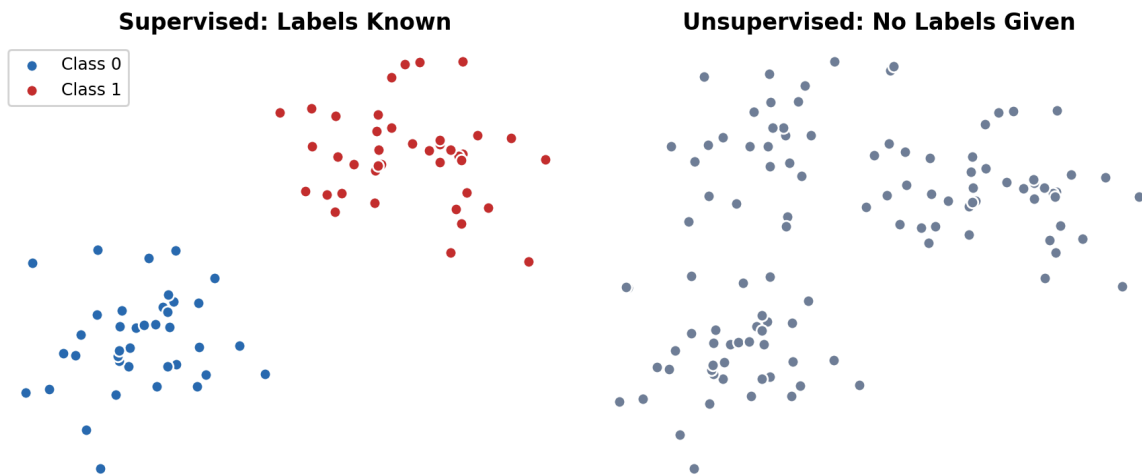
Pros

- No labeled data required — works directly on raw, cheap-to-collect data
- Can surface genuinely unexpected patterns a human wouldn't have thought to look for
- Useful as a preprocessing step even when the end goal is supervised (e.g., PCA before a classifier)

Cons

- No ground truth to check against — evaluation is harder and often more subjective
- Results can be difficult to interpret or validate (“is this cluster actually meaningful?”)
- Choosing the right number of clusters/components is often more art than science

3. Side by Side



Same underlying data — supervised learning knows the class of every point; unsupervised learning has to find any structure on its own.

Aspect	Supervised Learning	Unsupervised Learning
Input data	Labeled (x, y) pairs	Unlabeled (just x)
Goal	Predict a known target	Discover hidden structure
Ground truth exists?	Yes — you can check if you were right	No — no single correct answer to check against
Typical tasks	Classification, regression	Clustering, dimensionality reduction, association
Example algorithms	Linear/logistic regression, decision trees, random forest, XGBoost, SVM	k-Means, hierarchical clustering, PCA, DBSCAN, SVD/matrix factorization
Evaluation	Accuracy, precision/recall, F1, ROC-AUC, MAE/RMSE/R ²	Silhouette score, inertia, explained variance — often more subjective
Data cost	Expensive — labels often require manual annotation	Cheap — raw data is often already available
Example use case	Spam detection, churn prediction, price forecasting	Customer segmentation, anomaly detection, topic discovery

4. Quick Decision Guide

- **If** You have labeled data and a specific outcome to predict → use supervised learning.
- **If** You have no labels and want to explore/understand the data's structure → use unsupervised learning.
- **If** You have a little labeled data and a lot of unlabeled data → look into semi-supervised learning (not covered here, but worth knowing exists).
- **Note:** You're reducing a huge feature set before feeding it into a supervised model → that's unsupervised learning (PCA) used in service of a supervised pipeline — the two aren't mutually exclusive in practice.

5. Where This Connects to Your Roadmap

Everything through Stage 5 (linear/logistic regression, evaluation metrics) has been supervised learning — every dataset had a known target column. Your recommendation system project is a bridge case: collaborative filtering via SVD uses the unsupervised technique of matrix factorization (same core idea as PCA) even though it's applied to ratings data. k-Means and PCA later in Stage 4/6 are pure unsupervised learning — no target column at all, just discovering structure in the data itself.